LESSON 9.2

Analyze Graphs of Relationships Between Two Variables

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

REMEMBER

Analyze Graphs of Relationships Between Two Variables

Representing Two-Variable Relationships with Tables & Graphs. Formula: Table (x, y) → Graph Point (x, y) A table shows a relationship with numbers. A graph shows the same relationship with points. Reading either one tells you how the dependent variable responds to the independent variable.

1. Construct table: Put independent variable in column 1 (x) and dependent in column 2 (y).
2. Identify the rate of change or rule connecting x to y.
3. Plot the ordered pairs on the coordinate plane with x on horizontal axis and y on vertical

Words: graph *(gráfica)* · table of values *(tabla de valores)* · ordered pair *(par ordenado)* · axis *(eje)* · coordinates *(coordenadas)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each ordered pair (tickets, cost): on the graph, or not?

Groups: On the ticket graph · NOT on the graph

_____ (1, 45)

_____ (2, 90)

_____ (4, 180)

_____ (3, 120)

_____ (2, 100)

_____ (5, 200)

HINT Test each pair: does the cost equal 45 × the tickets?

2 **Complete the table. Show how you found each value.**

Tickets cost $45 each. Complete the table that feeds the graph.

Tickets (x)	Thinking	Total cost in dollars (y)
2		
5		
8		

HINT The independent variable (tickets) goes in column 1; the dependent (cost) responds in the last column.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Songs cost $2 each. Which is the independent variable?

(A) The number of songs bought
(B) The total cost
(C) The $2 price
(D) The music store

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

"The longer you run, the more calories you burn." Which is the dependent variable?

(A) Minutes run
(B) Calories burned
(C) The runner
(D) The distance

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A plant grows taller each day. Which sentence is correct?

(A) The number of days depends on height

(B) Height depends on the number of days

(C) Neither one depends on the other

(D) Both are independent variables

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?

SENTENCE STARTER The ordered pair ___ does / does not match because 45 × ___ = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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Set B · Version B · Second form · core practice

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LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

On the football ticket graph, what does the point (3, 135) represent?

(A) 3 tickets cost a total of $135
(B) 3 tickets cost $135 each
(C) 135 tickets cost a total of $3
(D) The game lasts 135 minutes

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

In any relationship between two quantities, the independent variable is the one that…

(A) you choose or control
(B) is always larger
(C) responds to the other
(D) is always graphed vertically

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Nakai opened a savings account with an initial deposit, or principal, of $750. The account earns 3.5% simple interest each year — interest is the amount earned for the use of the principal. A graph shows the interest Nakai has earned over a period of years, with points rising steadily. On Nakai's graph, which variable is plotted on the independent axis (x-axis)?

(A) years

(B) interest earned

(C) principal amount

(D) account balance

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Nakai opened a savings account with an initial deposit, or principal, of $750. The account earns 3.5% simple interest each year — interest is the amount earned for the use of the principal. A graph shows the interest Nakai has earned over a period of years, with points rising steadily. Why does the graph rise by the same amount every year instead of curving upward more steeply over time?

(A) Nakai earns compound interest, which grows faster each year

(B) The interest rate increases by 3.5% every year

(C) Nakai deposits an extra $26.25 into the account every year

(D) Nakai earns simple interest — the same $26.25 each year, calculated only on the original $750 principal

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

Nakai opened a savings account with an initial deposit, or principal, of $750. The account earns 3.5% simple interest each year — interest is the amount earned for the use of the principal. A graph shows the interest Nakai has earned over a period of years, with points rising steadily. A point on the graph is located at (4, x). Based on the interest rate, what value of x makes this point correct?

(A) $78.75

(B) $105

(C) $131.25

(D) $855



WORK

EXPLAIN YOUR THINKING

Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?

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